

Métodos Formais — Trabalho 1

Provas por Indução de Especificações Equacionais Recursivas

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1 Especificação das Funções

Para a realização das demonstrações, foram consideradas as seguintes especificações de funções recursivas sobre o tipo indutivo de listas $List(\tau)$:

$$\text{cat}([], ys) = ys \quad (\text{cateq1})$$

$$\text{cat}(x : xs, ys) = x : \text{cat}(xs, ys) \quad (\text{cateq2})$$

$$\text{reverso}([]) = [] \quad (\text{reveq1})$$

$$\text{reverso}(x : xs) = \text{cat}(\text{reverso}(xs), [x]) \quad (\text{reveq2})$$

2 Lema Auxiliar 1: Associatividade de cat

Enunciado:

$$\forall xs, ys, zs \in List(\tau) \cdot \text{cat}(xs, \text{cat}(ys, zs)) = \text{cat}(\text{cat}(xs, ys), zs)$$

Prova Formal Feita à Mão

$$\begin{aligned}
 & \text{cat} : \text{List} \langle T \rangle \times \text{List} \langle T \rangle \rightarrow \text{List} \langle T \rangle \\
 & \text{cat} ([], ys) = ys \quad (\text{cateq1}) \\
 & \text{cat} (x::xs, ys) = x::\text{cat}(xs, ys) \quad (\text{cateq2}) \\
 & \text{reverse} : \text{List} \langle T \rangle \rightarrow \text{List} \langle T \rangle \\
 & \text{reverse} ([]) = [] \quad (\text{reverse1}) \\
 & \text{reverse} (x::xs) = \text{cat} (\text{reverse}(xs), [x]) \quad (\text{reverse2})
 \end{aligned}$$

L1: $\forall xs, ys, zs \in \text{List} \langle T \rangle$
 $\text{cat}(xs, \text{cat}(ys, zs)) = \text{cat}(\text{cat}(xs, ys), zs)$
 $)$

Provar por indução em xs
 $\forall ys, zs \in \text{List} \langle T \rangle (\text{cat}(xs, \text{cat}(ys, zs)) = \text{cat}(\text{cat}(xs, ys), zs))$

Caso Base
 Seja $ys, zs \in \text{List} \langle T \rangle$ arbitrários
 provar $\text{cat}([], \text{cat}(ys, zs)) = \text{cat}(\text{cat}([], ys), zs)$
 $\text{cat}([], \text{cat}(ys, zs))$
 $= \text{cat}(ys, zs) \quad (\text{cateq1})$
 $= \text{cat}(\text{cat}([], ys), zs) \quad (\text{cateq1})$
 q.e.d.

Caso Indutivo
 Seja $xs \in \text{List} \langle T \rangle$, $x \in T$ arbitrários
 assumir $\text{HI} \forall ys, zs \in \text{List} \langle T \rangle (\text{cat}(xs, \text{cat}(ys, zs)) = \text{cat}(\text{cat}(xs, ys), zs))$
 provar $\forall ys, zs \in \text{List} \langle T \rangle (\text{cat}(x::xs, \text{cat}(ys, zs)) = \text{cat}(\text{cat}(x::xs, ys), zs))$
 $\text{cat}(x::xs, \text{cat}(ys, zs))$
 $= x::\text{cat}(xs, \text{cat}(ys, zs)) \quad (\text{cateq2})$
 $= x::\text{cat}(\text{cat}(xs, ys), zs) \quad (\text{HI})$
 $= \text{cat}(x::\text{cat}(xs, ys), zs) \quad (\text{cateq2})$
 $= \text{cat}(\text{cat}(x::xs, ys), zs) \quad \text{q.e.d.} \quad (\text{cateq2})$

Figura 1: Demonstração detalhada e passo a passo à mão para o Lema 1.

Prova em Isabelle / Isar

Listing 1: Código-fonte Isar para o Lema 1

```

1 lemma l1: "\<forall>ys zs :: 'a list . cat xs (cat ys zs) = cat (cat xs ys) zs"
2 proof (induction xs)
3   show "\<forall>ys zs :: 'a list . cat [] (cat ys zs) = cat (cat [] ys) zs"
4   proof (intro allI)
5     fix ys zs :: "'a list"
6     have "cat [] (cat ys zs) = cat ys zs" by (simp only: cateq1)
7     also have "... = cat (cat [] ys) zs" by (simp only: cateq1)
8     finally show "cat [] (cat ys zs) = cat (cat [] ys) zs" by (simp)
9   qed
10 next

```

```

11 fix xs::"'a list" and x::'a
12 assume HI:"\<forall> ys zs :: 'a list . cat xs (cat ys zs) = cat (cat xs ys) zs"
13 show "\<forall>ys zs :: 'a list . cat (x#xs) (cat ys zs) = cat (cat (x#xs) ys) zs"
14 proof (rule allI, rule allI)
15   fix ys zs :: "'a list"
16   have "cat (x#xs) (cat ys zs) = x # cat xs (cat ys zs)" by (simp only: cateq2)
17   also have "... = x # cat (cat xs ys) zs" by (simp only: HI)
18   also have "... = cat (x # cat xs ys) zs" by (simp only: cateq2)
19   also have "... = cat (cat (x#xs) ys) zs" by (simp only: cateq2)
20   finally show "cat (x#xs) (cat ys zs) = cat (cat (x#xs) ys) zs" by (simp)
21 qed
22 qed

```

3 Lema Auxiliar 2: Elemento Neutro à Direita de cat

Enunciado:

$$\forall xs \in List(\tau) \cdot cat(xs, []) = xs$$

Prova Formal Feita à Mão

L2: $\forall ns \in List(T) \ (cat(ns, []) = ns)$

Prova por indução em ns

$\forall ns \in List(T) \ (cat(ns, []) = ns)$

CASO BASE:

provar $cat([], []) = []$

$cat([], [])$

$= []$ (cateq1)

q.e.d.

CASO INDUTIVO:

Seja $ns \in List(T)$, $n \in T$ arbitrárias

assumir HI $cat(ns, []) = ns$

provar $\forall ns \in List(T), n \in T \ (cat(n:ns, []) = n:ns)$

$cat(n:ns, [])$

$= n:cat(ns, [])$ (cateq1)

$= n:ns$ (HF)

q.e.d.

Figura 2: Demonstração detalhada e passo a passo à mão para o Lema 2.

Prova em Isabelle / Isar

Listing 2: Código-fonte Isar para o Lema 2

```

1 lemma l2: "cat xs [] = xs"
2 proof(induction xs)
3   have "cat [] [] = []" by (simp only: cateq1)
4   then show "cat [] [] = []" by (simp)
5 next
6   fix xs::"'a list" and x::'a
7   assume HI: "cat xs [] = xs"
8   have "cat (x#xs) [] = x # cat xs []" by (simp only: cateq2)
9   have "x # cat xs [] = x # xs" by (simp only: HI)
10  then show "cat (x#xs) [] = x # xs" by (simp)
11 qed

```

4 Lema Auxiliar 3: Distribuição do reverso sobre cat

Enunciado:

$$\forall xs, ys \in List(\tau) \cdot \text{reverso}(\text{cat}(xs, ys)) = \text{cat}(\text{reverso}(ys), \text{reverso}(xs))$$

Prova Formal Feita à Mão

Nota: Utilizando "rev" para representar "reverso" por brevidade.

L3: $\forall xs, ys \in List(T) (\text{reverso}(\text{cat}(xs, ys)) = \text{cat}(\text{rev}(ys), \text{rev}(xs)))$
 prova por indução em xs
 $\forall xs, ys \in List(T) (\text{reverso}(\text{cat}(xs, ys)) = \text{cat}(\text{rev}(ys), \text{rev}(xs)))$
Caso BASE:
 seja $ys \in List(T)$ arbitrário
 provar $\forall xs \in List(T) (\text{rev}(\text{cat}([], ys)) = \text{cat}(\text{rev}(ys), \text{rev}([])))$
 $\text{rev}(\text{cat}([], ys))$
 $= \text{rev}(ys)$ (cateq 1)
 $= \text{cat}(\text{rev}(ys), [])$ (L2)
 $= \text{cat}(\text{rev}(ys), \text{rev}([]))$ (rev eq 1)

Figura 3

CASO INDUTIVO

seja $ns \in \text{List}(T)$, $n \in T$ arbitrários

assumir HI ($\text{rev}(\text{cat}(ns, ys)) = \text{cat}(\text{rev}(ys), \text{rev}(ns))$)

provar $\forall ys \in \text{List}(T)$ ($\text{rev}(\text{cat}(n:ns, ys)) = \text{cat}(\text{rev}(ys), \text{rev}(n:ns))$)

$$\begin{aligned}
& \text{rev}(\text{cat}(n:ns, ys)) \\
&= \text{rev}(n:\text{cat}(ns, ys)) && (\text{cateq 2}) \\
&= \text{cat}(\text{rev}(\text{cat}(ns, ys)), [n]) && (\text{reveq 2}) \\
&= \text{cat}(\text{cat}(\text{rev}(ys), \text{rev}(ns)), [n]) && (HI) \\
&= \text{cat}(\text{rev}(ys), \text{cat}(\text{rev}(ns), [n])) && (L1) \\
&= \text{cat}(\text{rev}(ys), \text{rev}(n:ns)) && (\text{reveq 2}) \\
&\quad q.e.d.
\end{aligned}$$

Figura 4: Demonstração detalhada e passo a passo à mão para o Lema 3.

Prova em Isabelle / Isar

Listing 3: Código-fonte Isar para o Lema 3

```

1 lemma l3: "\<forall>ys :: 'a list . reverso (cat xs ys) = cat (reverso ys) (reverso xs)"
2   proof (induction xs)
3     show "\<forall>ys :: 'a list . reverso (cat [] ys) = cat (reverso ys) (reverso [])"
4     proof (intro allI)
5       fix ys :: "'a list"
6       have "reverso (cat [] ys) = reverso ys" by (simp only: cateq1)
7       also have "... = cat (reverso ys) []" by (simp only: l2)
8       also have "... = cat (reverso ys) (reverso [])" by (simp only: reveq1)
9       finally show "reverso (cat [] ys) = cat (reverso ys) (reverso [])" by (simp)
10    qed
11  next
12    fix xs :: "'a list" and x::'a
13    assume HI: "\<forall>ys :: 'a list . reverso (cat xs ys) = cat (reverso ys) (reverso xs)"
14    show "\<forall>ys :: 'a list . reverso (cat (x#xs) ys) = cat (reverso ys) (reverso (x#xs))"
15    proof (intro allI)
16      fix ys :: "'a list"
17      have "reverso (cat (x#xs) ys) = reverso (x#(cat xs ys))" by (simp only: cateq2)
18      also have "... = cat (reverso (cat xs ys)) [x]" by (simp only: reveq2)
19      also have "... = cat (cat (reverso ys) (reverso xs)) [x]" by (simp only: HI)
20      also have "... = cat (reverso ys) (cat (reverso xs) [x])" by (simp only: l1)
21      also have "... = cat (reverso ys) (reverso (x#xs))" by (simp only: reveq2)
22      finally show "reverso (cat (x#xs) ys) = cat (reverso ys) (reverso (x#xs))" by (simp)
23    qed
24  qed

```

5 Teorema Principal: Involução do reverso

Enunciado:

$$\forall xs \in List(\tau) \cdot \text{reverso}(\text{reverso}(xs)) = xs$$

Prova Formal Feita à Mão

Nota: Utilizando "rev" para representar "reverso" por brevidade.

The image shows a handwritten proof on a light brown background. At the top, there is a horizontal line with two double slashes in the center. Below this, the theorem is stated: $\forall ns \in List\langle T \rangle (rev(rev(ns)) = ns)$. Below the theorem, it says "prova por indução em ns". The proof is divided into two cases: "CASO BASE:" and "CASO INDUTIVO:". The base case shows that $rev(rev([])) = []$ by simplifying the expression and using the definition of rev on an empty list. The inductive case starts with "seja $ns \in List\langle T \rangle, n \in T$ " and "assumir $H \vdash rev(rev(ns)) = ns$ ". It then shows that $rev(rev(n:ns)) = n:ns$ by using the definition of rev on a cons cell and the inductive hypothesis. The steps are: $rev(rev(n:ns)) = rev(concat(rev(ns), [n]))$ (by definition), $= concat(rev([n]), rev(rev(ns)))$ (by definition), and $= concat(rev([n]), ns)$ (by the inductive hypothesis H). The final result is $n:ns$.

$\forall ns \in List\langle T \rangle (rev(rev(ns)) = ns)$
prova por indução em ns

CASO BASE:
provar $rev(rev([])) = []$
 $rev(rev([]))$
 $= rev([])$ (rev eq 1)
 $= []$ (rev eq 1)
q.e.d.

CASO INDUTIVO:
seja $ns \in List\langle T \rangle, n \in T$
assumir $H \vdash rev(rev(ns)) = ns$
provar $\forall ns \in List\langle T \rangle, n \in T (rev(rev(n:ns)) = n:ns)$
 $rev(rev(n:ns))$
 $= rev(concat(rev(ns), [n]))$ (rev eq 2)
 $= concat(rev([n]), rev(rev(ns)))$ (L3)
 $= concat(rev([n]), ns)$ (HI)

Figura 5

$= \text{cat}(\text{rev}(n:L], xs)$	(def. List(T))
$= \text{cat}(\text{cat}(\text{rev}(L], [n]), xs)$	(reveq2)
$= \text{cat}(\text{cat}(L], [n]), xs)$	(reveq1)
$= \text{cat}([n], xs)$	(cateq1)
$= \text{cat}(n:L], xs)$	(def. List(T))
$= n:\text{cat}(L], xs)$	(cateq2)
$= n:xs$	(cateq1)
q.e.d.	

Figura 6: Demonstração detalhada e passo a passo à mão para o Teorema Principal.

Prova em Isabelle / Isar

Listing 4: Código-fonte Isar para o Teorema Principal

```

1 theorem t1: "reverso (reverso xs) = xs"
2 proof (induction xs)
3   have "reverso (reverso []) = reverso []" by (simp only: reveq1)
4   also have "reverso [] = []" by (simp only: reveq1)
5   then show "reverso (reverso []) = []" by (simp)
6 next
7   fix xs :: "'a list" and x :: 'a
8   assume HI: "reverso (reverso xs) = xs"
9   have "reverso (reverso (x#xs)) = reverso (cat (reverso xs) [x])" by (simp only: reveq2)
10  also have "... = cat (reverso [x]) (reverso (reverso xs))" by (simp only: l3)
11  also have "... = cat (reverso (x#[])) xs" by (simp only: HI) (* transf. [x] \<rightarrow> x
#[] nesse passo *)
12  also have "... = cat (cat (reverso []) [x]) xs" by (simp only: reveq2)
13  also have "... = cat (cat [] [x]) xs" by (simp only: reveq1)
14  also have "... = cat (x#[]) xs" by (simp only: cateq1) (* transf. [x] \<rightarrow> x#[]
nesse passo *)
15  also have "... = x # (cat [] xs)" by (simp only: cateq2)
16  also have "... = x # xs" by (simp only: cateq1)
17  finally show "reverso (reverso (x#xs)) = x # xs" by (simp)
18 qed

```